Phu Nguyen

My first dataset, Telco Customer Churn, got rejected because it was already clean and there was nothing for me to fix. That taught me the point of this assignment is the cleaning itself, so I switched to the Stroke Prediction dataset, which is much messier.

The biggest thing I learned is that missing data doesn't always look missing. The bmi column had real blanks, but smoking_status used the word "Unknown" for about a third of the patients. I had to turn that into a real missing value before I could fill it.

I also learned why you split the data before doing anything else. I split 70/30 in Python first and only worked on the training part, so the test data stayed untouched.

I learned the difference between encoding, normalization, and scaling. Encoding turns text into numbers, like Male = 0 and Female = 1. Normalization changes the shape of a column, and I used a log transform on glucose to lower its skew from 0.92 to 0.44. Scaling changes the range, and I squeezed age, glucose, and bmi into 0 to 1. When I first opened the final CSV the numbers looked weird, but I figured out they were just scaled values and I could convert them back. I made a second, readable version of the file so people can see real ages and words.

Something I'd do differently is test my choices more. Filling 30% of smoking_status with the most common value is a guess, and capping glucose may have removed real risk signals.